USE CASES & OVERALL SYSTEM DESIGN

Inventory Management System (IMS)
for Thrifted & Branded Surplus Clothing
(with Item Approval Workflow)

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1. Purpose

This document describes the system use cases and overall design of the IMS. It follows the documented workflow from item intake, physical review and approval, sale processing, stock monitoring, and reporting.

2. Use Case Summary

ID	Use Case	Actor	Expected Result
UC-01	Log In	Staff / Reviewer / Admin	User receives access based on assigned role.
UC-02	Add New Item	Staff	Item is saved automatically as Pending.
UC-03	View Added Items and Status	Staff	Staff sees the current status of items they added.
UC-04	Review Pending Item	Reviewer	Reviewer records the result of the physical inspection.
UC-05	Approve Item	Reviewer	Pending item becomes In Stock.
UC-06	Reject Item	Reviewer	Item returns to Pending with a mandatory rejection reason.
UC-07	Process Sale	Staff	In Stock item becomes Sold and stock is reduced.
UC-08	Search Inventory	Admin	Matching inventory records are displayed.
UC-09	Monitor Stock Levels	Admin	Stock levels and low-stock alerts are visible.
UC-10	Generate Status Reports	Admin	Status report is generated from logged data.

2.1 Use Case Diagram

The use case diagram shows how Staff, Reviewer, and Admin interact with the IMS and its main functions.

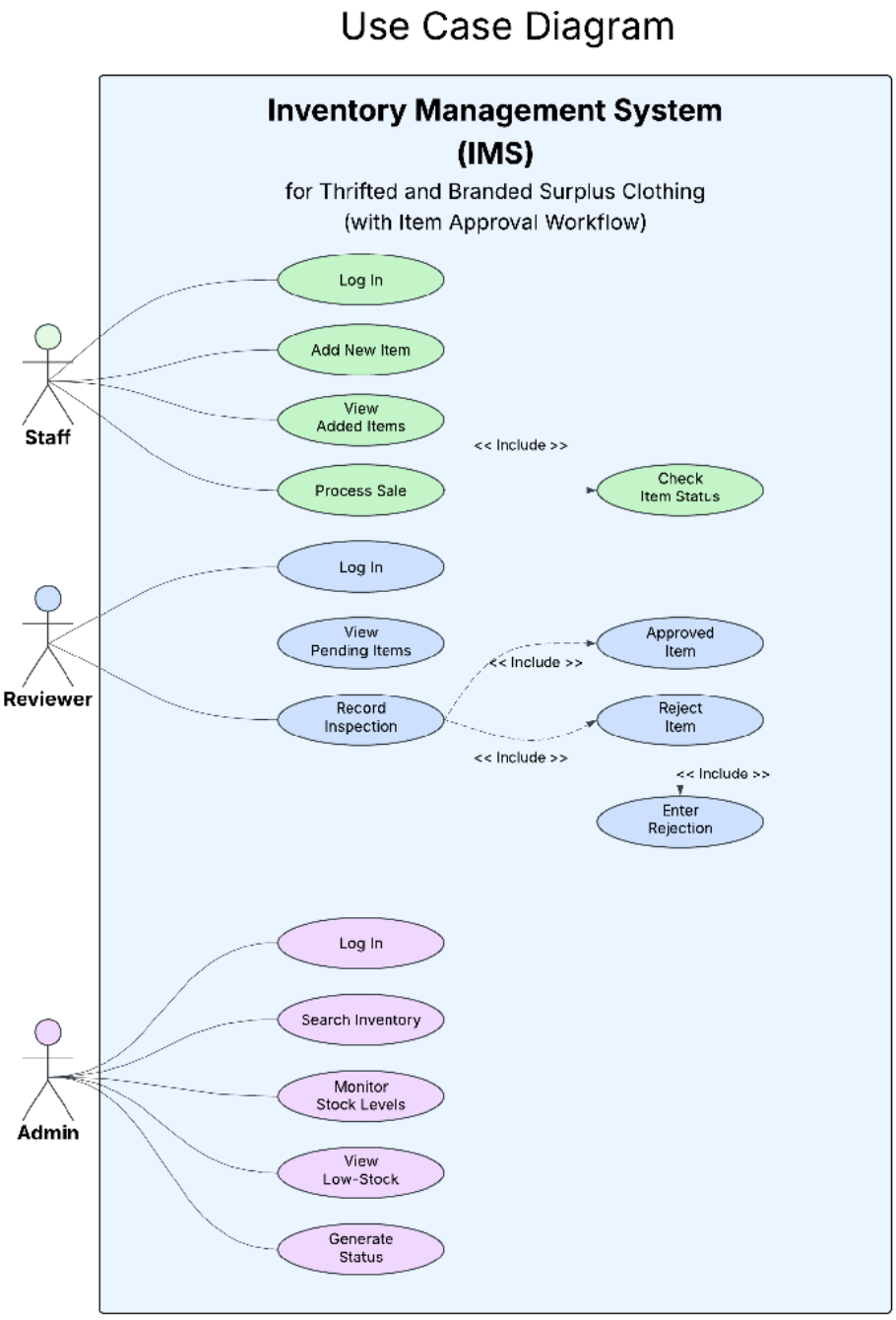


Figure 1.

Use Case Diagram of the Inventory Management System

3. Detailed Use Cases

UC-02 – Add New Item

Field	Description
Actor	Staff
Precondition	Staff is logged in.
Main Flow	Staff opens Add Item; enters brand, size, condition, price, and photo; submits the form; system validates and saves the item as Pending.
Postcondition	A Pending item is created.

UC-04 – Review Pending Item

Field	Description
Actor	Reviewer
Precondition	A Pending item exists.
Main Flow	Reviewer opens the Pending Items queue, selects an item, physically checks quality, condition, tags, and authenticity, then records Approve or Reject.
Postcondition	The decision is processed and logged.

UC-05 – Approve Item

Field	Description
Actor	Reviewer
Precondition	Item is Pending and has been inspected.
Main Flow	Reviewer selects Approve; system changes Pending to In Stock and logs the change.
Postcondition	Item is In Stock and can be sold.

UC-06 – Reject Item

Field	Description
Actor	Reviewer
Precondition	Item is Pending and has been inspected.
Main Flow	Reviewer selects Reject; system requires a reason; Reviewer enters it; system returns/keeps the item as Pending and records the reason.
Postcondition	Item remains Pending with rejection reason.

UC-07 – Process Sale

Field	Description
Actor	Staff
Precondition	Item is In Stock.
Main Flow	Staff selects the item and processes the sale; Status Guard verifies In Stock status; system changes it to Sold, reduces stock, and logs the change.
Postcondition	Item is Sold.

UC-08 – Search Inventory

Field	Description
Actor	Admin
Precondition	Admin is logged in.
Main Flow	Admin searches by brand, size, condition, status, or price; system returns matching records.
Postcondition	Matching records are displayed.

UC-09 – Monitor Stock Levels

Field	Description
Actor	Admin
Precondition	Admin is logged in.
Main Flow	Admin opens the inventory dashboard and reviews stock levels and low-stock indicators.
Postcondition	Stock condition can be monitored.

UC-10 – Generate Status Reports

Field	Description
Actor	Admin
Precondition	Admin is logged in.
Main Flow	Admin opens Reports and requests a status report; system compiles Pending, In Stock, and Sold data, rejection reasons, and status history.
Postcondition	Report is displayed.

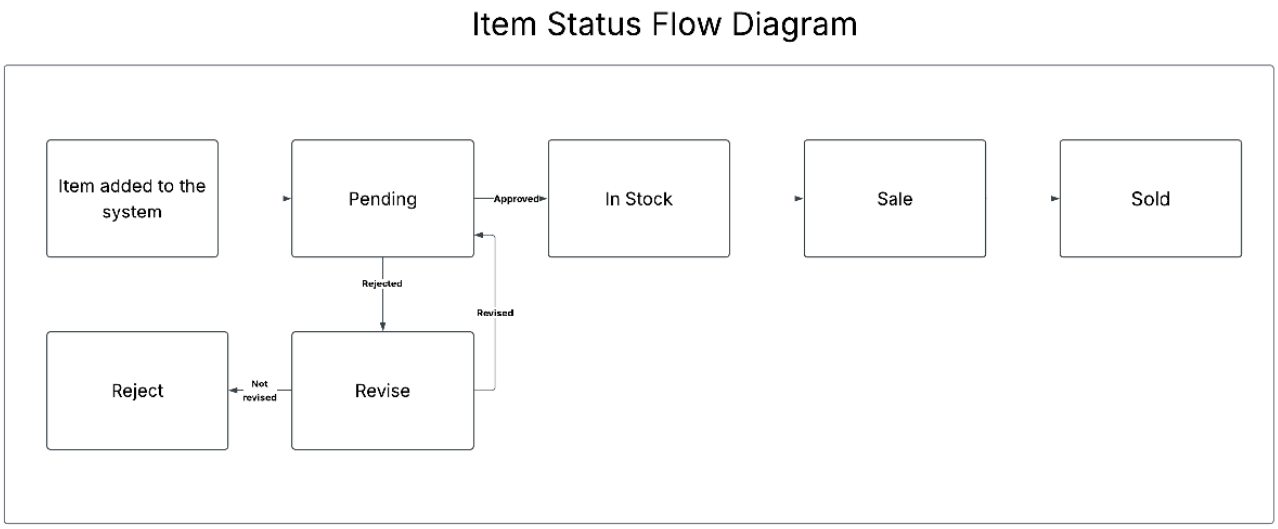
4. Overall System Design

The system is organized around role-based access and the garment status lifecycle. The system records the result of the physical inspection but does not perform the inspection itself.

4.1 Main System Areas

Module / Area	Main User	Purpose
Authentication & Role Access	All roles	Validates login and restricts functions according to role.
Item Intake	Staff	Captures garment information and creates the item as Pending.
Approval Workflow	Reviewer	Displays Pending items and records Approve/Reject decisions.
Sales & Stock	Staff	Processes sales only for In Stock items and updates stock.
Inventory Monitoring	Admin	Supports inventory search and stock/low-stock monitoring.
Reporting	Admin	Generates status reports from inventory and status history.
Status Logging	System-wide	Records each status transition with user and timestamp.

4.2 Item Status Flow



4.3 End-to-End Workflow

1. Staff logs in and submits a new item with brand, size, condition, price, and photo.
2. The system validates the input and saves the item as Pending.
3. The item appears in the Pending Items queue and cannot be sold.
4. Reviewer physically inspects the garment and records Approve or Reject.
5. Approve changes the item to In Stock. Reject requires a reason and returns the item to Pending.
6. Every decision is written to the status log.
7. Staff processes a sale only when the item is In Stock.
8. A successful sale changes the item to Sold and reduces stock.
9. Admin searches inventory, monitors stock and low-stock alerts, and generates status reports.

4.4 Planned Interface Screens

Screen	Primary User	Purpose
Login	All roles	Authentication and role-based access.
Add Item Form	Staff	Record garment details and photo.
Pending Items Queue	Reviewer	Review Pending items and Approve/Reject.
Inventory List / Dashboard	Admin	Search and monitor stock.
Sale / Checkout	Staff	Process an In Stock item sale.
Reports	Admin	View status reports.

4.5 Core Data Areas

The Development Plan explicitly identifies Items, Users, and Status Logs for database design. The source documents do not provide a complete field-level database schema, so no additional tables or attributes are assumed here.

Data Area	Purpose
Items	Stores inventory item information and current status.
Users	Supports role-based accounts.
Status Logs	Stores item status changes, previous/new status, user, and timestamp.